

IoT Platform

COAP Communication Protocol

Version V1.0.0

2021-12-09

Revision Record

Version	Date	Description
1.0.0	2021-12-09	Initial draft

1. Introduction

This protocol is the communication protocol between the NB-IoT water meter and the data acquisition platform. The connection method uses the LwM2M protocol. LwM2M is an IoT communication protocol based on the COAP protocol. On top of COAP, LwM2M defines standardized interfaces and objects, making communication between IoT devices and the IoT Platform more concise and standardized.

Communication Process

- The IoT water meter uploads data packets containing information such as cumulative flow and valve status to the IoT Platform, for example Telecom AEP or China Mobile OneNet.
- After the IoT Platform receives the data packet, it actively pushes the packet to a third-party application system. The third-party system parses and stores the data according to this protocol specification.
- The third-party application system calls the IoT Platform API to preset commands such as valve opening/closing and time calibration to the IoT Platform. When the IoT water meter comes online, the platform delivers the command to the meter.

2. Frame Structure

- This communication protocol refers to the CJ/T 188-2004 industry standard.
- All protocol characters are represented in hexadecimal HEX format.

2.1 Frame Format

Name	Code	Length (bytes)
Frame start symbol	68H	1
Meter type	T	1
Address field	A0-A6	7
Control code	C	1
Data length	L	1
Data identifier	DI0, DI1	2
Sequence number	SER	1
Data	DATA	N
Checksum	CS	1
End character	16H	1

2.2 Meter Type and Code

Meter Type	Code
Water meter	10H
Heat meter	20H

2.3 Address Field

A0-A6 consist of seven bytes. Each byte is in 2-digit BCD format. The address length is 14 decimal digits. The lower address is transmitted first and the higher address is transmitted later.

2.4 Control Code

Bit	Meaning
D7	0 = communication frame sent by the platform; 1 = communication frame sent by the water meter.
D6-D0	Refer to the specific protocol parsing rules.

2.5 Data Length

Data length = number of data identifier bytes + sequence number bytes + data field bytes. The value is represented in hexadecimal.

2.6 Data

The actual information of the meter.

2.7 Checksum

One byte. It is the accumulated sum of all bytes from the frame start symbol up to the byte before the checksum. Overflow beyond FFH is ignored.

3. Application Protocol

3.1 Water Meter Data Upload: Data Identifier 9097

Control code: 81H

Data length: ACH

Data identifier: 90H, 97H

Sequence number: SER

Frame Format

68H	10	A0-A6	81H	ACH	90H	97H	SER	DATA	CS	16H
-----	----	-------	-----	-----	-----	-----	-----	------	----	-----

68H 10 A0-A6 81H ACH 90H 97H SER DATA CS 16H

DATA Field

Data Content	Data Format	Bytes
Current cumulative flow unit	Refer to Data Format 1	1
Current cumulative flow	BCD, low byte first, high byte later	4
Settlement-day cumulative flow unit	Refer to Data Format 1	1
Settlement-day cumulative flow	BCD, low byte first, high byte later	4
Reverse cumulative flow unit	Refer to Data Format 1	1
Reverse cumulative flow	BCD, low byte first, high byte later	4
Remaining flow unit	Refer to Data Format 1	1
Remaining flow	BCD, low byte first, high byte later	4
Instantaneous flow-rate unit	Refer to Data Format 2	1
Instantaneous flow rate	BCD, low byte first, high byte later	4
Current temperature	Refer to Data Format 3	2
Current pressure	Refer to Data Format 4	2
Ultrasonic signal	HEX	2
Meter internal time	Refer to Data Format 5	7
Meter status code	Refer to Data Format 6	2
Signal strength	HEX - 0xFFFF	2
Signal quality	HEX	2
Transmission count	HEX	2
ICCID	BCD, high byte first, low byte later	10
IMEI	BCD, high byte first, low byte later	8
Upload flag	HEX	1

Timing scheme	BCD	4
48 freeze-data cutoff time, 00:00:00	BCD	1
48 freeze-data cutoff flow, 00:00:00 flow	BCD, low byte first, high byte later	4
Previous day 23:30:00 negative increment	HEX	2
Previous day 23:00:00 negative increment	HEX	2
...	...	...
Previous day 00:30:00 negative increment	HEX	2

3.1.1 Detailed Parsing Example

Data:

```
68107545000013082181AC9097002B606620002B308001002B00000000
00000000035000000092270000000000461404260821200002FFB4001200
0B8986111925301747443008648230479880500104FFFFFF00606620000000
00000000000000000000000000000000000000000000000000000000000000
00000000000000F6E07941D1011621B9F09EB060E079E0D35048759ED00
000D6F07760B7D0A0017FB0F6515C20D0B16450D0B068614D2079500B3
2CD8E616
```

Protocol category: 819097

Item	Parsed Value	Protocol Data
Frame start symbol	68	68
Meter type	10	10
Meter address	21081300004575	75450000130821
Control code	81	81
Data length	172	AC
Data identifier	9097	9097
SER	00	00
Cumulative flow unit	0.001 m³	2B
Cumulative flow	206.66	60662000
Settlement-day cumulative flow unit	0.001 m³	2B
Settlement-day cumulative flow	18.03	30800100
Reverse cumulative flow unit	0.001 m³	2B
Reverse cumulative flow	0	00000000
Remaining flow unit	00	00
Remaining flow	0	00000000
Flow-rate unit	1 x 10 ⁻⁴ m³/h	35
Flow rate	0.0000	00000000
Temperature	27.92 °C	922700
Pressure	0.00 MPa	0000
Ultrasonic signal	0000	0000
Meter internal time	2021/08/26 04:14:46	46140426082120
Meter status code	0002	0002
Signal strength	-76	FFB4
Signal quality	18	0012
Transmission count	11	000B
ICCID	89861119253017474430	89861119253017474430
IMEI	864823047988050	0864823047988050
Trigger flag	01	01
Timing scheme	04FFFFFF	04FFFFFF
24-hour freeze cutoff time	00:00	00
24-hour freeze cutoff flow	206.66	60662000
Time point 23:30:00 negative increment	0	0000
Time point 23:00:00 negative increment	0	0000
Time point 22:30:00 negative increment	0	0000
Time point 22:00:00 negative increment	0	0000
Time point 21:30:00 negative increment	0	0000
Time point 21:00:00 negative increment	0	0000
Time point 20:30:00 negative increment	0	0000
Time point 20:00:00 negative increment	0	0000
Time point 19:30:00 negative increment	0	0000
Time point 19:00:00 negative increment	0	0000
Time point 18:30:00 negative increment	0	0000
Time point 18:00:00 negative increment	0	0000
Time point 17:30:00 negative increment	0	0000
Time point 17:00:00 negative increment	0	0000
Time point 16:30:00 negative increment	0	0000
Time point 16:00:00 negative increment	0	0000
Time point 15:30:00 negative increment	0	0000
Time point 15:00:00 negative increment	0	0000
Time point 14:30:00 negative increment	0	0000

Time point 14:00:00 negative increment	0	0000
Time point 13:30:00 negative increment	3.95	0F6E
Time point 13:00:00 negative increment	1.94	0794
Time point 12:30:00 negative increment	7.44	1D10
Time point 12:00:00 negative increment	4.45	1162
Time point 11:30:00 negative increment	7.071	1B9F
Time point 11:00:00 negative increment	2.539	09EB
Time point 10:30:00 negative increment	1.55	060E
Time point 10:00:00 negative increment	1.95	079E
Time point 09:30:00 negative increment	3.381	0D35
Time point 09:00:00 negative increment	1.159	0487
Time point 08:30:00 negative increment	23.021	59ED
Time point 08:00:00 negative increment	0	0000
Time point 07:30:00 negative increment	3.439	0D6F
Time point 07:00:00 negative increment	1.91	0776
Time point 06:30:00 negative increment	2.941	0B7D
Time point 06:00:00 negative increment	2.56	0A00
Time point 05:30:00 negative increment	6.139	17FB
Time point 05:00:00 negative increment	3.941	0F65
Time point 04:30:00 negative increment	5.57	15C2
Time point 04:00:00 negative increment	3.339	0D0B
Time point 03:30:00 negative increment	5.701	1645
Time point 03:00:00 negative increment	3.339	0D0B
Time point 02:30:00 negative increment	1.67	0686
Time point 02:00:00 negative increment	5.33	14D2
Time point 01:30:00 negative increment	1.941	0795
Time point 01:00:00 negative increment	0.179	00B3
Time point 00:30:00 negative increment	11.48	2CD8
Checksum	E6	E6
End symbol	16	16

3.2 Platform-Issued Valve Control Command

Note: For platform-issued commands, the meter type value is fixed as 20H.

Control code: 04H

Data length: 04H

Data identifier: A0H, 17H

Sequence number: SER

Frame Format

Command	Frame
Open valve	68H 20 A0-A6 04H 04H A0H 17H SER 55 CS 16H
Close valve	68H 20 A0-A6 04H 04H A0H 17H SER 99 CS 16H

Detailed Parsing Example

Command	Example Frame
Open valve	68 20 75 45 00 00 13 08 21 04 04 A0 17 00 55 92 16
Close valve	68 20 75 45 00 00 13 08 21 04 04 A0 17 00 99 D6 16

3.3 Time Calibration + Timing Scheme

Control code: 04H

Data length: 11H

Data identifier: A1H, 18H

Sequence number: SER

Frame Format

68H 20 A0-A6 04H 11H A1H 18H SER DATA CS 16H

DATA Field

Data Content	Data Format	Bytes
Time difference	Refer to the detailed parsing example	7
Timing scheme	Refer to the detailed parsing example	5
Reserved	Reserved	2

Detailed Parsing Example

68 20 AA AA AA AA AA AA 04 11 A1 18 00 00 05 04 30 06
20 00 01 04 08 15 20 00 00 52 16

Time Difference Calibration

- BYTE1 to BYTE3: 00 05 04. Time difference: 4 hours, 5 minutes, 0 seconds.
- BYTE4 to BYTE6: 30 06 20. Date: no addition or subtraction; assign directly when BYTE7 is not 00.
- BYTE7: 00 means the date is not calibrated. The meter internal date remains unchanged.
- For time updating, seconds are not configured. Minutes and hours are added directly according to the difference value: meter internal time + time difference.
- If minutes exceed 60, hours +1. If hours exceed 24, no carry is required.
- Time calibration is relatively complicated. If time calibration is not required, set all time-difference bytes to 00.

68 20 AA AA AA AA AA AA 04 11 A1 18 00 00 00 00 00 00
00 01 04 08 15 20 00 00 3E 16

Timing Scheme

- 01 04 08 15 20:
- 01 means fixed-time upload.
- 04 08 15 20 means the meter reports data at 04:00, 08:00, 15:00, and 20:00 according to the meter internal time.

4. Data Format Description

4.1 Data Format 1: Cumulative Flow Unit

Code	Unit
29H	0.01 L
2AH	0.1 L
2BH	L
2CH	0.01 m ³
2DH	0.1 m ³
2EH	1 m ³

4.2 Data Format 2: Instantaneous Flow Unit

Code	Unit
32H	1 x 10 ⁻⁷ m ³ /h
33H	1 x 10 ⁻⁶ m ³ /h
34H	1 x 10 ⁻⁵ m ³ /h
35H	1 x 10 ⁻⁴ m ³ /h
36H	1 x 10 ⁻³ m ³ /h
37H	1 x 10 ⁻² m ³ /h

4.3 Data Format 3: Temperature Value

Byte	High Nibble / D7-D4	Low Nibble / D3-D0
BYTE 1	BCD tenths digit	BCD hundredths digit
BYTE 2	BCD tens digit	BCD ones digit

4.4 Data Format 4: Pressure Value

Byte	High Nibble / D7-D4	Low Nibble / D3-D0
BYTE 1	BCD hundredths digit	BCD thousandths digit
BYTE 2	BCD ones digit	BCD tenths digit

4.5 Data Format 5: Time

Byte	Format
BYTE 1	BCD seconds
BYTE 2	BCD minutes
BYTE 3	BCD hours
BYTE 4	BCD day
BYTE 5	BCD month
BYTE 6	BCD year, low byte
BYTE 7	BCD year, high byte

4.6 Data Format 6: Water Meter Status

Water Meter Status BYTE 1

Bits	Definition	Description
D1D0	Valve status	00 = open; 01 = closed; 11 = abnormal
D2	Battery voltage	0 = normal; 1 = low battery voltage alarm
D3-D7	Reserved	Reserved

Water Meter Status BYTE 2

Bit	Definition	Description
D0	Battery level alarm	0 = normal; 1 = alarm
D1	Empty pipe alarm	0 = normal; 1 = alarm
D2	Backflow alarm	0 = normal; 1 = alarm
D3	Over-range alarm	0 = normal; 1 = alarm
D4	Water temperature alarm	0 = normal; 1 = alarm
D5	EE alarm	0 = normal; 1 = alarm
D6-D7	Reserved	Reserved

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